Chemical-Reaction-Controlled Phase Separated Drops: Formation, Size Selection, and Coarsening

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Phase separation under nonequilibrium conditions is exploited by biological cells to organize their cytoplasm but remains poorly understood as a physical phenomenon. Here, we study a ternary fluid model in which phase-separating molecules can be converted into soluble molecules, and vice versa, via chemical reactions. We elucidate using analytical and simulation methods how drop size, formation, and coarsening can be controlled by the chemical reaction rates, and categorize the qualitative behavior of the system into distinct regimes. Ostwald ripening arrest occurs above critical reaction rates, demonstrating that this transition belongs entirely to the nonequilibrium regime. Our model is a minimal representation of the cell cytoplasm.

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Phase separation is a ubiquitous phenomenon in our physical world, ranging from cloud formation to oil drop formation in water [1]. Recently, it has also been realized that phase separation is exploited in the cell cytoplasmic organization in the formation of nonmembrane bound organelles called ribonucleoprotein (RNP) granules [2,3]. RNP granules are a diverse set of structures that play important roles in the functioning of the cell, from RNA processing and stress response [4,5] to cell division [6] and germ line specification [7,8]. However, the mechanisms enabling the rapid and controlled assembly and disassembly of RNP granules have only begun to be investigated. Chemical reactions, e.g., in the form of adenosine triphosphate (ATP)-driven enzymatic reactions that convert one protein state to another (e.g., unphosphorylated to phosphorylated) are prime candidates for the cell to manifest controlled phase separation. For instance, such a scheme has been proposed as a means to induce localized phase separation in the *Caenorhabditis elegans* embryo [9–11], and to organize the centrosomes prior to cell division [6]. However, the physics of nonequilibrium phase separation driven by chemical reactions has only started to be investigated. For instance, nonequilibrium processes have been discussed in the context of lipid domains in plasma membranes [12–14]. More recently, it has been realized that although in equilibrium phase separation a multidrop, finite system will invariably coarsen to a single condensed drop via Ostwald ripening, nonequilibrium chemical reactions can arrest this ripening process completely in a binary fluid [15,16]. Here, we categorize comprehensively and under general conditions how unimolecular reactions that convert a two-state molecule between a phase-separating state and a soluble state can control drop formation, coarsening, and size selection. We achieve this by generalizing and improving upon the assumptions adopted in Ref. [16]. Specifically, we analyze the regimes of large drops and non-negligible supersaturation, include the presence of cytosol by going beyond the binary fluid restriction, and allow for arbitrary equilibrium concentrations inside and outside drops. Our model is arguably the minimal model relevant to the mechanism of chemical-reaction-controlled phase separation in the cell cytoplasm.

Our ternary mixture consists of two molecular states, one phase separating (P) and one soluble (S), plus the solvent or cytosol (C). States P and S can be converted into each other by the chemical reactions

$$P \underset{h}{\overset{k}{\rightleftharpoons}} S, \tag{1}$$

where k and h are the reaction rate constants. The nonequilibrium nature of these reactions lies in the fact that both reaction rates are independent of the local concentrations and thus have to be driven by free-energy consumption. In the context of the cell, these reactions can be, e.g., ATP-driven post-transcriptional protein modifications [17] that affect protein phase-separating behavior. For example, the phase separation of intrinsically disordered proteins can be regulated via their phosphorylation or dephosphorylation [18,19].

At equilibrium ($k, h = 0$), a finite system will inevitably coarsen via Ostwald ripening [20] and drop coalescence [21]. Here, we assume that drop diffusion is negligible, so we will focus exclusively on the Ostwald ripening. In the cell context, this is motivated by the strong suppression of macromolecular diffusion in the cell cytoplasm [22]. Ostwald ripening results from two effects: (1) the Gibbs-Thomson relation dictating that for a drop of size R, the concentrations of solute inside and outside the drop next to the interface are $\hat{P}_{\rm in}$ and $\hat{P}_{\rm out}(1 + l_c/R)$, respectively, where

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l_c is the capillary length and $\hat{P}_{\mathrm{in,out}}$ are the equilibrium phase coexistence concentrations [see Fig. 1(a)], and (2) the concentration profile of the solute in the dilute phase is given by the steady-state solution to the diffusion equation (the quasistatic assumption). These two effects combined lead to a diffusive flux of solute from small drops to big drops [20].

When chemical reactions are switched on, we assume that local thermal equilibrium remains valid so that the interface boundary conditions for P are unchanged [23]. S is considered inert to phase separation in the sense that its concentration profile is continuous across the interface [26]. In addition, we assume that the concentration profiles inside and outside the drops are given by

$$\frac{\partial P_{\mathrm{in,out}}}{\partial t} = D\nabla^2 P_{\mathrm{in,out}} - kP_{\mathrm{in,out}} + hS_{\mathrm{in,out}}, \tag{2}$$

$$\frac{\partial S_{\mathrm{in,out}}}{\partial t} = D\nabla^2 S_{\mathrm{in,out}} + kP_{\mathrm{in,out}} - hS_{\mathrm{in,out}}, \tag{3}$$

where $P_{\mathrm{in,out}}$ and $S_{\mathrm{in,out}}$ denote the concentration profiles of P and S inside and outside drops with subscripts "in" and "out," respectively. For simplicity, we assume the same diffusion coefficient D for both species and in both phases.

To see why Ostwald ripening can be arrested in our ternary mixture, we will now provide an intuitive argument based on a similar consideration for binary mixtures [16]. We consider a homogeneous system of total solute concentration $\phi = P_{\mathrm{tot}} + S_{\mathrm{tot}}$, where P_{tot} (S_{tot}) is the total concentration of P (S) in the system. If the supersaturation $\Delta = P_{\mathrm{tot}} - \hat{P}_{\mathrm{out}}$ is positive, drops can be nucleated, initiating phase separation [Fig. 1(a)]. At small Δ the drop density is low and drops only interact with the far-field concentration. We focus only on the early growth regime so that the supersaturation Δ remains close to $P_{\mathrm{tot}} - \hat{P}_{\mathrm{out}}$: for a drop of radius R, the diffusive profile leads to an influx of P into the drop at the rate [20]

$$4\pi DR\left(\Delta - \frac{\hat{P}_{\mathrm{out}} l_c}{R}\right). \tag{4}$$

At the same time, the chemical reactions inside the drop lead to a depletion of P at the rate

$$\frac{4\pi R^3}{3} k\hat{P}_{\mathrm{in}}. \tag{5}$$

As R increases, the depletion rate will eventually surpass the influx from the medium, so that the balance between Eqs. (4) and (5) leads to a steady-state radius. In the limit of large R so that we can ignore the term $\hat{P}_{\mathrm{out}} l_c/R$ (but R still small such that $\Delta \approx P_{\mathrm{tot}} - \hat{P}_{\mathrm{out}}$), the steady-state R is

$$\sqrt{\frac{3D\Delta}{k\hat{P}_{\mathrm{in}}}}. \tag{6}$$

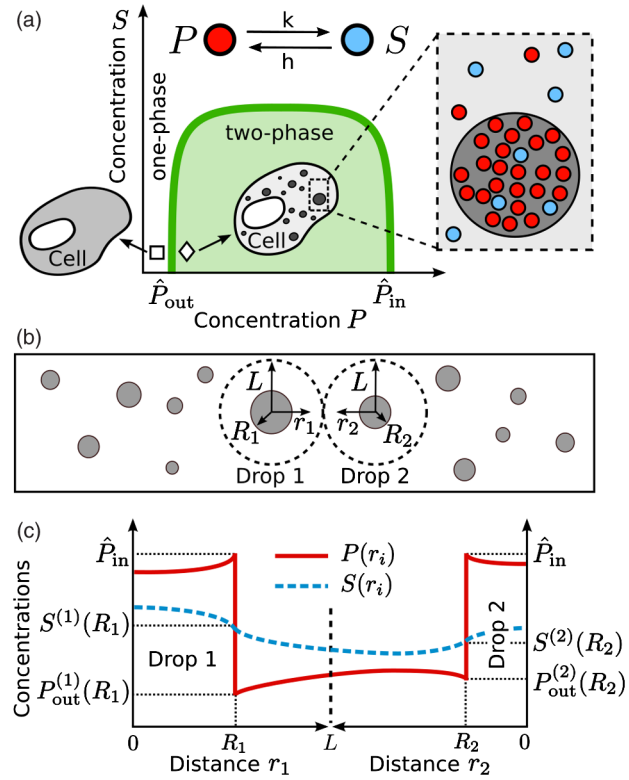

FIG. 1. Model of cytoplasmic phase separation. (a) The cell cytoplasm is modeled by a ternary fluid composed of phase-separating (P) and soluble (S) molecular states, and other cytoplasmic components (C). Chemical reactions convert P into S at the rate k, and S into P at the rate h [Eq. (1)]. At equilibrium ($k = h = 0$), the system is well mixed (square symbol) if the concentrations of P and S lie outside the phase boundary (green line in the phase diagram), and the system phase separates otherwise (diamond symbol). In the latter case, we assume that S does not phase separate and remains homogeneous. (b) A multidrop system with drop number density ρ is studied by considering two interacting subsystems ($i = 1,\ 2$) of radius $L = [3/(4\pi\rho)]^{1/3}$, each having a drop of radius R_i in their center. (c) Schematics of the concentration profiles of P and S in the subsystems when chemical reactions are present [$k, h > 0$, Eqs. (8) and (9)]. At the subsystems boundaries ($r_i = L$) the profiles and their derivatives are matched by assumption [Eq. (7)].

We shall see that this regime in fact corresponds to the upper bound of stable R in a multidrop system and in a specific regime (upper dotted line in Fig. 3).

In our argument above, we have neglected the reverse reaction $S \to P$ and the effect of the chemical reactions on the diffusive profiles, which, as we shall see, can significantly change the system behavior. We will now incorporate these effects into our analysis. We will also consider arbitrary supersaturations so that drops may be close to each other. As a result, a far-field concentration may not exist, rendering the Lifshitz-Slyozov theory [20] inapplicable. Consider a multidrop system such that drops are on average a distance $2L$ apart where L is of the order $\rho^{-1/3}$

with ρ being the drop number density. For simplicity, we will first focus on two spherical subsystems of radius L, each having a spherical drop in their center [Fig. 1(b)]. We assume that the concentrations and their gradients at the boundaries of the two subsystems match [Fig. 1(c)]. The rationale for this approximation is that in a multidrop system, the actual boundary conditions are influenced by many neighboring drops and we treat these fluctuating boundary conditions in a *mean-field* manner by assuming spherical symmetry around the drops. In other words, the concentration outside a drop depends only on the distance from the drop center. Moreover, it is assumed that the two-drop system is stable (unstable) if the full multidrop system is stable (unstable). The validity of this approximation will be verified later using Monte Carlo simulations. The corresponding boundary conditions, besides the Gibbs-Thomson relation at the drop interfaces, are

$$P^{(1)}_{\rm out}(L) = P^{(2)}_{\rm out}(L), \qquad \nabla_{\mathbf{r}_1} P^{(1)}_{\rm out}|_L = -\nabla_{\mathbf{r}_2} P^{(2)}_{\rm out}|_L, \quad (7)$$

and the same apply to $S^{(i)}_{\rm in,out}$. The subscript $i = 1, 2$ denotes the drop index. Note that we use two different coordinate systems r_1 and r_2, each having their respective drop center as the origin [Fig. 1(b)].

Using the quasistatic approximation as in the equilibrium case, the steady-state concentration profiles of this two-drop system with radii R_1 and R_2, such that $R_1 \approx R_2$, are [24]

$$P^{(i)}_{\rm in}(r_i) = \frac{I_i h}{k+h} + H^{(i)}_{\rm in} \frac{R_i}{r_i} \frac{\sinh(r_i/\xi)}{\sinh(R_i/\xi)}, \quad (8)$$

$$P^{(i)}_{\rm out}(r_i) = \frac{Oh}{k+h} + H^{(i)}_{\rm out} \frac{R_i}{r_i} (A_i e^{r_i/\xi} + B_i e^{-r_i/\xi}). \quad (9)$$

In the above, A_i, B_i are independent of r_i and are given in Ref. [24], I_i, O denote the combined concentration $P + S$ inside and outside the ith drop, respectively, and are also independent of r_i. Furthermore, $H^{(i)}_{\rm in} \equiv \hat{P}_{\rm in} - I_i h/(k+h)$, $H^{(i)}_{\rm out} \equiv P^{(i)}_{\rm out}(R_i) - Oh/(k+h)$, and $\xi \equiv \sqrt{D/(k+h)}$ is a concentration gradient length scale. The S profiles are given by $S^{(i)}(r_i) = I_i - P^{(i)}_{\rm in}(r_i)$ for $r_i < R_i$ and $S^{(i)}(r_i) = O - P^{(i)}_{\rm out}(r_i)$ for $r_i > R_i$. Note that, generally, O is independent of r only when $R_1 = R_2$, which we have assumed to be true here as we will focus on the case $R_1 \approx R_2$.

The volumetric growth rate of the ith drop in this two-drop system is [6,24]

$$G_i(R_i, R_j) = \frac{4\pi D R_i^2}{\hat{P}_{\rm in} - P^{(i)}_{\rm out}(R_i)} \left(\frac{dP^{(i)}_{\rm out}}{dr_i}\bigg|_{R_i^+} - \frac{dP^{(i)}_{\rm in}}{dr_i}\bigg|_{R_i^-} \right), \quad (10)$$

and we recall that $P^{(i)}_{\rm out}(R_i)$ is given by $\hat{P}^{(i)}_{\rm out}(1 + l_c/R_i)$ via the Gibbs-Thomson relation. Given the drop growth rate above we can study the steady-state drop radius R^* at which the two drops of the same size are in the steady state ($G_i = 0$).

We can also analyze the steady-state stability by calculating the drop growth rates upon perturbing their size: $R_1 \mapsto R^* + \epsilon$ and $R_2 \mapsto R^* - \epsilon$. Performing a linear stability analysis, we take $\epsilon \ll R^*$ and expand the growth rate with respect to ϵ:

$$G_1(R_1, R_2) = g_0(R^*) + g_1(R^*)\epsilon + \mathcal{O}(\epsilon). \quad (11)$$

Solving for $g_0(R^*) = 0$ gives the steady-state drop radius R^* and the sign of $g_1(R^*)$ indicates the stability of the system: coarsening will occur if $g_1 > 0$, while the system is stable if $g_1 < 0$. Using the profiles Eqs. (8) and (9), we find

$$g_0(R^*) = \frac{4\pi D R^*}{\hat{P}_{\rm in} - P_{\rm out}(R^*)} \left\{ H_{\rm in} \left(1 - \frac{R^*}{\xi} \coth \frac{R^*}{\xi} \right) - H_{\rm out} \left[A \left(1 - \frac{R^*}{\xi} \right) e^{R^*/\xi} + B \left(1 + \frac{R^*}{\xi} \right) e^{-R^*/\xi} \right] \right\}, \quad (12)$$

with $H_{\rm in,out} \equiv H^{(i)}_{\rm in,out}(R_1 = R_2 = R^*)$ and A and B are functions of R^*/ξ and L/ξ [24]. The expression of $g_1(R^*)$ is more complicated and is shown in Ref. [24].

The surface plot in Fig. 2 shows, for a fixed backward reaction rate h, the region of existence of the steady-state

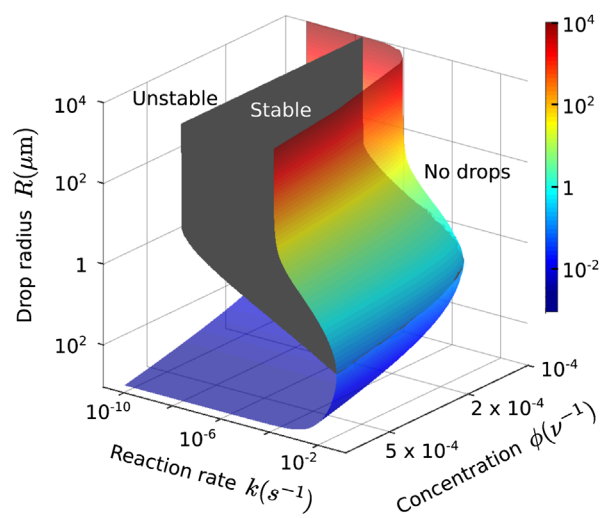

FIG. 2. The stability of a multidrop system at fixed backward reaction rate h. The region of existence of a steady-state radius R^* [solution of $g_0(R^*) = 0$, Eq. (12)] is controlled by the forward reaction rate k and the total solute concentration ϕ. In the region enclosed by the colored outer surface, R^* exists and depends on the drop number density ρ, which is not fixed in this figure. The steady state is stable [$g_1(R^*) < 0$, Eq. (11)] above the black inner surface, and unstable [$g_1(R^*) > 0$] below this surface. Parameters are $h = 10^{-2}$ s^{-1} $l_c = 10^{-2}$ μm, $D = 1$ μm^2 s^{-1}, $\hat{P}_{\rm in} = 5 \times 10^{-2}$ ν^{-1}, $\hat{P}_{\rm out} = 10^{-4}$ ν^{-1}, where ν is the molecular volume of P and S and can be chosen arbitrarily.

radius R^*, delimited by the colored outer surface. Above the black inner surface, the system consists of stable monodisperse drops whose sizes are controlled by the rate k, the solute concentration ϕ, and the drop number density ρ (not fixed in Fig. 2). Outside the stable region but still within the outer surface, the monodisperse system is in an unstable steady state and drops coarsen via Ostwald ripening. Outside the outer surface, drops always shrink.

Interestingly, there are qualitative changes in the system behavior as k varies as shown in Fig. 3, which describes multidrop stability at fixed solute concentration ϕ and fixed backward rate h. When $k < k_l$ (blue arrow), the system is in the Lifshitz-Slyozov regime and coarsens (upward arrow), while for $k_l < k < k_u$ (green arrow), the system can be stable (gray region), with coexisting drops of radius determined by ρ. In other words, k_l is the critical rate beyond which Ostwald ripening is arrested. Between k_u and k_c (black arrow), the system can also be stable, but with an upper bound on the radius. Beyond k_c, no drops can exist in the system as all drops evaporate (downward arrows).

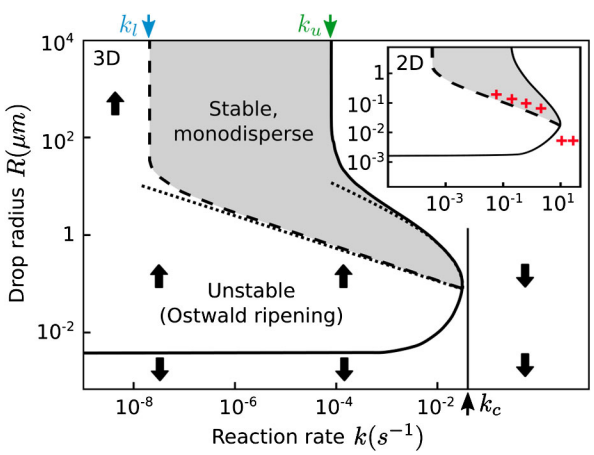

FIG. 3. Stability diagram of a multidrop system at fixed backward rate h and fixed solute concentration ϕ. A steady-state drop radius R^* exists in the region enclosed by the continuous curve and depends on the rate k and the drop number density ρ. Outside this region no steady states exist and drops dissolve (downward arrows). The lower part of this line represents the smallest possible drop, or nucleus. Outside the gray region but still within the continuous line, the steady state is unstable causing the average radius to increase (upward arrows). The stability-instability boundary $[g_1(R^*) = 0]$ is shown with a dashed line. There is a good agreement between our analytical calculation and the numerical solutions for k_l, k_u, k_c [Eqs. (16) and (17)]. The analytical expressions for the upper bound radius R_u [Eq. (13)] and the stability-instability boundary R_l [Eq. (15)] in the small drop regime ($R^* \ll \xi$) are shown by the dotted lines. Parameters are $\phi = 5 \times 10^{-4}\nu^{-1}$, $\hat{P}_{\rm in} = 10^{-1}\nu^{-1}$, and the rest are as in Fig. 2. Inset: Comparison between 2D Monte Carlo simulations and 2D numerical solutions to the linear stability analysis. Simulation data are shown in red; note that the two rightmost crosses represent the size of the lattice site ($\sim 10^{-2}$ μm); i.e., there are no drops in the system. The region $k \approx k_l$ is also investigated in Ref. [24]. See Ref. [24] for the corresponding analysis in 2D and simulation details.

So far, our calculation has been based on our two-drop system with the mean-field matching assumption at the system boundaries. To test this assumption, we perform Monte Carlo simulations of our ternary model on a 2D lattice with multiple drops to detect the stability-instability boundary (black inner surface in Fig. 2 and dashed curve in Fig. 3) and compare the results with our predictions (see Ref. [24] for simulation details.) The good agreements are shown in the inset of Fig. 3 and in Ref. [24].

We will now explain analytically the salient features of the stability diagram by focusing on distinct limits in the small supersaturation limit.

Upper bound on drop radius.—We have seen that if $k_u < k < k_c$, there is an upper bound on the drop radius. We focus here on the regime $R^* \ll \xi$, which we will see is indeed the case when $k \gg k_u$. We first analyze the limit of small drop number density ρ so that the distance between drops is large: $L \gg \xi$. By expanding g_0 with respect to the small parameters R^*/ξ and ξ/L, we seek the set of R^* such that the solutions to $g_0(R^*) = 0$ cease to exist. We find that the expression of this boundary is [24]

$$R_u = \sqrt{\frac{3D\left[\frac{h\phi}{k+h} - \hat{P}_{\rm out}\left(1 + \frac{l_c}{R_u}\right)\right]}{k\hat{P}_{\rm in}}} \tag{13}$$

$$\underset{R_u \gg l_c}{\simeq} \sqrt{\frac{3D\left(\frac{h\phi}{k+h} - \hat{P}_{\rm out}\right)}{k\hat{P}_{\rm in}}}, \tag{14}$$

which is indicated by the upper dotted line in Fig. 3. We have thus recovered the result Eq. (6) obtained by intuitive arguments since $P_{\rm tot} = h\phi/(k+h)$ [24].

Stability-instability boundary.—For the stability-instability boundary, we remain in the regime of small R^*, but now focus on large ρ so that $L \ll \xi$. In this case, the small parameters are R^*/ξ and L/ξ. By expanding g_0 and g_1 around these small parameters, we solve for the steady state $g_0(R^*) = 0$ and then seek the boundary of stability by looking at $g_1(R^*) = 0$. The functional form of this boundary is [24]

$$R_l = \left(\frac{3Dl_c\hat{P}_{\rm out}}{2k\hat{P}_{\rm in}}\right)^{1/3}, \tag{15}$$

which is indicated by the lower dotted line in Fig. 3. We note that similar scaling laws to Eqs. (14) and (15) have previously been found for binary mixtures [15,16], as well as in the context of domain formation in biomembranes [12].

Critical reaction rate k_c.—The rate k_c beyond which drops dissolve is the solution of $R_u(k_c) = 0$ and is maximally bounded as follow [24]:

$$k_c < \min\left[\frac{\phi - \hat{P}_{\rm out}}{\hat{P}_{\rm out}} h\,;\ \frac{4D(\phi - \hat{P}_{\rm out})^3}{9 l_c^2 \hat{P}_{\rm in} \hat{P}_{\rm out}^2}\right]. \qquad (16)$$

Note that $k > (\phi - \hat{P}_{\rm out})/\hat{P}_{\rm out} h$ corresponds to the situation where the conversion $P \to S$ is so strong that the system is outside the equilibrium phase-separating region [$P_{\rm tot} < \hat{P}_{\rm out}$; see Fig. 1(a)].

Lower and upper critical rates (k_l *and* k_u).—Here, we focus on the large drop limit so that the small parameters are ξ/R^* and ξ/L (since $L > R^*$). By expanding g_0, g_1 with respect to these two small parameters, we solve again for $g_0 = 0$ and investigate the corresponding stability by looking at g_1. Specifically, we find [24]

$$k_l = \frac{2 l_c \hat{P}_{\rm out}}{D^{1/2} \hat{P}_{\rm in}} h^{3/2}, \qquad k_u = \frac{2(\phi - \hat{P}_{\rm out})}{\hat{P}_{\rm in}} h, \qquad (17)$$

where k_l is the transition rate from the stable to the unstable regime, and k_u is the rate beyond which drop radii have an upper bound. Thus, the transition to the nonequilibrium regime, namely, the arrest of Ostwald ripening, occurs at nonzero reaction rate (k_l). This behavior has never been reported before in this system.

Finally, we note that given the richness of the system behavior, the generic features of the stability diagram can vary according to h, which we have explored in Ref. [24] and in the context of cellular response to environmental stresses [27].

In summary, we have studied a phase-separating ternary fluid mixture with chemically active drops. We have categorized the qualitative behavior of the system into distinct regimes based on the reaction rates using a combination of analytical, numerical, and simulation methods. Our work is of direct importance to cytoplasmic organization, and is also relevant to the control of emulsions in the engineering setting. To make our theory applicable in more general settings, interesting directions include the incorporation of drop coalescence into our coarsening picture, which can be especially relevant at large drop number density, the study of potential shape instabilities in chemically active drops [28], and the generalization of our formalism to many-component mixtures [29,30].

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